

Cognitive function in women with breast cancer receiving adjuvant chemotherapy and healthy controls

Noriko Ando-Tanabe · Yumi Iwamitsu · Masaru Kuranami · Shigemi Okazaki · Hiroe Yasuda · Yuki Nakatani · Kenji Yamamoto · Masahiko Watanabe · Hitoshi Miyaoka

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Abstract

Background The aim of this study was to evaluate the impact of chemotherapy and psychological distress on cognitive function in patients with breast cancer receiving chemotherapy compared to healthy controls.

Methods Eighteen women with non-metastatic breast cancer who have been exposed to chemotherapy were evaluated with a battery of neuropsychological and psychological tests before (time 1) and 1 month after chemotherapy (time 2). Twenty healthy women were assessed at matched intervals.

Results The relationship between cognitive change and mood change from time 1 to time 2 was compared between breast cancer patients and healthy controls. Independent

t tests revealed no differences in change scores for each neuropsychological test and psychological questionnaire between the chemotherapy and control groups. However, in the chemotherapy group, change scores for verbal memory test and executive function test were significantly and negatively correlated with change score for Hospital Anxiety and Depression Scale (HADS)-D ($r \leq -0.498$, $p < 0.05$). Furthermore, change scores for the verbal memory test and processing function test were also significantly and negatively correlated with change score for HADS-A ($r \leq -0.503$, $p < 0.05$). The multiple regression model for the processing function test accounted for a significant amount of variance (model adjusted $R^2 = 0.157$, $p < 0.05$), where the standard partial regression coefficient for HADS-A change score was statistically significant ($\beta = -0.441$, $p < 0.01$), whereas the standard partial regression coefficients for the group and for the HADS-D change score were -0.297 ($p = 0.071$) and 0.026 ($p = 0.868$), respectively.

Conclusions Higher psychological distress was associated with poor cognitive function in patients receiving chemotherapy in this study.

Keywords Breast cancer · Chemotherapy · Cognitive function · Oncology · Psychological distress

N. Ando-Tanabe (✉)

Department of Medical Psychology, Pharmaceutical Education Research Center, Kitasato University School of Pharmacy, 5-9-1 Shirokane, Minato-ku, Tokyo 108-8641, Japan
e-mail: andon@pharm.kitasato-u.ac.jp

Y. Iwamitsu · S. Okazaki · H. Yasuda · Y. Nakatani
Department of Medical Psychology, Graduate School of Medical Sciences, Kitasato University, 1-15-1 Kitasato, Minami-ku, Sagami-hara, Kanagawa 252-0373, Japan

M. Kuranami · M. Watanabe
Department of Surgery, Kitasato University School of Medicine, 1-15-1 Kitasato, Minami-ku, Sagami-hara, Kanagawa 252-0373, Japan

S. Okazaki
Higashiyamato General Hospital, 1-13-12 Nangai, Higashiyamato, Tokyo 207-0014, Japan

K. Yamamoto · H. Miyaoka
Department of Psychiatry, Kitasato University School of Medicine, 1-15-1 Kitasato, Minami-ku, Sagami-hara, Kanagawa 252-0373, Japan

Introduction

Over the last couple of decades, a growing number of reports have suggested that chemotherapy leads to cognitive impairment, which is referred to as “chemo-brain (chemo-fog)” in the lay literature. According to cross-sectional studies that evaluated cognitive functioning in women undergoing breast cancer treatment, cognitive

impairment was more pronounced in patients who received chemotherapy than those who received other therapy or healthy women, both during chemotherapy and more than 10 years after completion of chemotherapy [1–8]. However, other studies found no evidence to support a significant influence of chemotherapy on cognitive functioning [9, 10]. In recent years, several longitudinal or observational studies have been conducted which provide evidence for chemotherapy-induced cognitive change. Although research methodology regarding case–control setting and time points of cognitive function assessment varied in each study, many of these studies suggested that chemotherapy was associated with changes in cognitive function in breast cancer patients [11–14]. On the other hand, other studies reported that chemotherapy was not associated with changes in cognitive function in breast cancer patients [15–18].

When interpreting results of the longitudinal or observational studies, the magnitude of practice effects should be considered in a certain type of neuropsychological testing [19, 20]. In fact, previous research on longitudinal study designs revealed that a total or part of cognitive test performance of breast cancer patients improved following chemotherapy or toward the end of treatment compared to before chemotherapy (at baseline) [11, 12, 16]. Some researchers examined practice effects in order to understand chemotherapy-induced cognitive impairment [16, 21]. Cognitive impairment assessed with a longitudinal or observational methodology should be discussed with appropriate attention to practice effects that occur in repeated neuropsychological assessments.

Confounding results from recent studies have complicated our understanding of the impact of chemotherapy on cognitive functioning. Some breast cancer patients demonstrate evidence of cognitive impairment prior to chemotherapy or after surgery following adjuvant therapy [21–23]. Previous studies reported that cognitive impairment in breast cancer patients who did not receive chemotherapy was related to age [24, 25], education level [14, 24, 25], and psychological distress in patients [23]. Furthermore, cancer diagnosis and treatment can cause psychological distress in patients, which may interfere with their cognitive functioning [26]. In fact, previous studies revealed that psychological distress was significantly related to overall/partial cognitive impairment in breast cancer patients [3, 23]. On the other hand, some studies suggested that psychological distress was not a factor affecting patient cognitive function [16, 18, 22, 25, 27].

To date, only a limited number of studies regarding “chemo-brain” have been conducted with prospective or case–control methodology. The aim of the present study was to investigate the effects of chemotherapy on cognitive function in women undergoing breast cancer treatment relative to healthy controls using a prospective and

observational study design. The impacts of chemotherapy on cognitive functioning of women with breast cancer were evaluated at two time points. Furthermore, we assessed both cognitive function and psychological distress (depressive symptom and anxiety) before and 4 weeks after completion of chemotherapy, focusing on changes in cognitive function as well as mood in patients relative to healthy individuals.

Patients and methods

Patients and healthy controls

A total of 26 patients with primary breast cancer, who were drawn from the population of patients initiating chemotherapy with adriamycin and cyclophosphamide (AC)/epirubicin and cyclophosphamide (EC) followed by paclitaxel/docetaxel regimen at Kitasato University Hospital Department of Surgery between July 2006 and June 2009, were recruited for this study. Patients who met the following criteria were considered eligible to participate in the study: (1) women aged 18 years or older, (2) patients who had not undergone treatment for severe mental illness or other severe diseases, (3) patients who had not received previous treatment for any cancer, (4) patients who had not received previous treatment for any organic brain disorder, (5) patients who were able to read and speak Japanese. Of the 26 study participants, 1 patient did not complete the first round of neuropsychological test sets, and 7 patients did not complete the second round, including 5 patients who refused to continue study participation because of time constraints or their perceived burden of completing the second round of test sets, 1 patient who did not reply to our recall, and 1 patient who discontinued chemotherapy after one cycle of AC treatment. In the end, 18 patients were subject to the following analysis of neuropsychological assessments and questionnaire scores, and 94.4 % of these patients were reassessed between 4 and 8 weeks after the completion of chemotherapy.

The control group consisted of healthy volunteers, such as nurses who worked at clinics, clerical workers, and graduate students. Healthy controls were recruited for study participation if they met the following criteria in addition to patient criteria: (1) they were not undergoing treatment for any cancer, (2) they did not know much about neuropsychological tests. Of 22 eligible controls who were recruited to participate in this study between January 2008 and March 2008, 2 dropped out; 1 had no time to complete the test and 1 felt it was a burden to complete the test and refused to continue study participation. In the end, the control group consisted of 20 healthy individuals.

Procedures

All patients and healthy controls who met the eligibility criteria were recruited for study participation. After provision of written informed consent, baseline assessment was undertaken on the day of recruitment or at another time before the first chemotherapy treatment (time 1; T1). A follow-up assessment was conducted at about 4 weeks after the final chemotherapy session (about 8–10 months after T1 assessment in the control group, time 2; T2), and at about 6 months after the final chemotherapy session (about 14–16 months in the control group, time 3; T3). In this study, we analyzed T1 and T2 data (T3 data will be reported elsewhere).

At each assessment, participants completed a self-report questionnaire and were administered neuropsychological evaluations by the researchers with appropriate training and supervision. Testing took approximately 1 h and 20 min. Study procedures were approved by the Institutional Review Board of Kitasato University School of Medicine and Hospital.

Neuropsychological tests

Logical memory I and II from the Wechsler Memory Scale-Revised (WMS-R): participants were presented with two short paragraphs individually, and following each presentation, asked to recall the paragraph from memory (logical memory I). Delayed recall was obtained after 30 min (logical memory II). Scores were generated on the basis of the scaled score of the Japanese version of WMS-R.

Verbal paired associates I and II from the WMS-R: these tests consisted of eight word pairs (four “easy” pairs and four “hard” pairs), which were read to participants who took a memory trial following each reading. Scoring was based on the first three trials, although participants who did not learn all the pairs by the third trial were given up to three more trials (verbal paired associates I). Approximately 30 min later, a single recall trial was given (verbal paired associates II). Both easy and hard pairs received the same one-point value for each correct response.

Visual reproduction I and II from the WMS-R: participants were individually presented with four novel designs for 10 s, and following each presentation, asked to reproduce the design from memory (visual reproduction I). Delayed recall was obtained after 30 min (visual reproduction II).

Digit span from the WMS-R: digit span forward measures how many bits of information a person can attend to at one time in a task requiring repetition of a random series of digits presented at a rate of one digit per second. Digit span backward requires sustained directed attention

to mentally track and reverse a random series of digits. Scores were obtained as the number of digits repeated correctly in forward and backward sequences before two failed trials.

Digit symbol from the Wechsler Adult Intelligence Scale-Revised (WAIS-R): participants were asked to match numbers with geometric marks according to a designated code. The total number of correct responses in 90 s was used.

Block design from the WAIS-R: participants were asked to arrange blocks with all white sides, all red sides, and red and white sides according to a presented pattern. Scores were generated on the basis of the scaled score of the Japanese version of WAIS-R.

Verbal Fluency Test (VFT): VFT consisted of the letter fluency test and category fluency test. Participants orally generated words beginning with a target letter over a series of three (shi, i, re) within 60 s per letter in the letter fluency test, and words from certain semantic categories, such as animal naming, types of transportation, and fruit naming within 60 s per category in the category fluency test. The total number of words was used.

Keio version Wisconsin Card Sorting Test (KWCST): participants were presented with four stimulus cards with symbols that differ in shape, number, and color, as well as a pack of response cards, and asked to select each response card under one of the four stimulus cards. We used the KWCST that has a reduced number of cards compared to the original version.

Estimated Intelligence Quotient (EIQ): in order to estimate baseline intelligence level, four subsets of WAIS-R, such as information, picture completion, similarities, and digit symbol, were administered.

Verbal memory score was calculated by adding double the score of logical memory I to the score of verbal paired associates I.

Delayed recall score was calculated by adding the score of logical memory II, double the score of verbal paired associates II, and the score of visual reproduction II.

Psychological test

The Hospital Anxiety and Depression Scale (HADS): HADS is a self-rating questionnaire that measures current levels of depression and anxiety symptoms [28]. We used the Japanese version of HADS [29] in this study.

Demographic and clinical data

Demographic data were obtained from participants via a self-report measure. Information regarding patient medical variables was obtained from patient medical records with informed consent.

Data analysis

First, demographic and clinical characteristics of patients who received chemotherapy, including participants and non-participants of this study, and healthy individuals in the control group were examined using the independent *t* test to verify whether there were differences in characteristics between the chemotherapy group and the control group. Furthermore, neuropsychological test scores at T1 were compared between the chemotherapy and control groups, although scores were not corrected for age. We did not statistically compare participants and non-participants of this study in the chemotherapy group, as the sample size of the non-participant group was small. Second, the change score for each neuropsychological test and psychological questionnaire was calculated by subtracting T1 score from T2 score, and change scores for each of the tests and questionnaire were also compared between the chemotherapy and control groups using an independent Student's *t* test. Third, to explore the relationship between change scores of neuropsychological variables and those of psychological variables, Pearson's correlation coefficients were calculated. Finally, multiple regression analyses were performed to determine possible predictors of change scores of neuropsychological tests in study participants. Group (chemotherapy or control group), change score for HADS-D, and change score for HADS-A were entered as independent variables to identify possible predictors of change score of each neuropsychological test. Using dummy variables in the regression model, we assigned a value of 1 (control group = 0) to the chemotherapy group. $p < 0.05$ (two-tailed) was considered statistically significant. Post hoc power analyses were conducted to calculate observed power values ($1 - \beta$) for significant chemotherapy effects ($\alpha = 0.05$). Statistical analyses were performed using SPSS 19.0 (SPSS Inc, Chicago) and G*Power 3 [30].

Results

Characteristics of participants and non-participants in the chemotherapy group and the control group

The characteristics of study participants and non-participants in the chemotherapy group and the control group are shown in Tables 1 and 2.

Regarding baseline data of neuropsychological tests, an independent *t* test was used to compare each test score at T1 for the chemotherapy and control groups. The results revealed that the digit symbol score for the chemotherapy group (mean $\pm$ SD = 63.5 ± 14.4) was lower than that of the control group [71.9 ± 10.5 ; $t(36) = -2.066$, $p < 0.05$, $r = 0.33$]. Moreover, scores of the letter fluency test and

category fluency test for the chemotherapy group (27.6 ± 6.0 and 42.2 ± 6.3) were also lower than those of the control group [33.4 ± 7.1 and 50.6 ± 8.2 ; $t(36) = -2.715$ and -3.517 , $p < 0.05$, $r = 0.41$ and 0.51]. Other neuropsychological test scores were not significantly different between the chemotherapy and control groups.

Comparison of change scores between the chemotherapy and control groups during the period between T1 and T2

The change scores for each neuropsychological test and HADS (depressive symptom and anxiety) for the chemotherapy and control groups during the period between T1 and T2 are shown in Tables 3 and 4. The results of independent *t* tests revealed no significant differences in each change score between the chemotherapy and control groups.

Association between change in neuropsychological test performance and change in anxiety/depression scale

Pearson's correlation coefficients between change score for each neuropsychological test and that of HADS-D and HADS-A for each group are shown in Table 5. In the chemotherapy group, change scores for the logical memory I, logical memory II, verbal memory (containing logical memory I), and category fluency test were significantly and negatively correlated with change score for HADS-D ($r \leq -0.498$, $p < 0.05$). Furthermore, change scores for verbal paired associates II and digit symbol were also significantly and negatively correlated with change score for HADS-A ($r \leq -0.503$, $p < 0.05$) in the chemotherapy group. On the other hand, the change score for CA was significantly and positively correlated with the change score for HADS-A ($r = 0.516$, $p < 0.05$) in the chemotherapy group. Observed power values ($1 - \beta$) for significant results are shown in Table 5.

Effect of chemotherapy and change in mood on cognitive function

Group, HADS-D change score, and HADS-A change score were entered as independent variables for each change score of neuropsychological tests using multiple regression analysis. Most analyses did not identify significant models; however, a significant model regarding digit symbol was obtained by multiple regression analysis ($R^2 = 0.229$, adjusted $R^2 = 0.157$, $p < 0.05$, $1 - \beta = 0.75$), where the standard partial regression coefficient for HADS-A change score was statistically significant ($\beta = -0.441$, $p < 0.01$). In this model, the standard partial regression coefficients for the group and HADS-D change score were -0.297 ($p = 0.071$) and 0.026 ($p = 0.868$), respectively.

Table 1 Characteristics and medical information of participants and non-participants

Examined for cognitive assessment at	Chemotherapy group participants	Chemotherapy group non-participants	Control group	Chemo participants vs. control participants	
	T1 and T2 (<i>n</i> = 18)	T1 only (<i>n</i> = 6)	T1 and T2 (<i>n</i> = 20)	<i>t</i> score	<i>p</i> value
Age at T1 (in years)					
Mean (SD)	51.2 (10.7)	60.2 (11.4)	47.0 (7.5)	1.403	0.169
Range	33–69	46–76	32–67		
Education					
Junior high	1	0	0		
High school	9	3	0		
College	3	0	10		
University	4	1	5		
Graduate school	0	0	4		
Not reported	1	2	1		
Marital status					
Never married	0	0	2		
Married	17	2	18		
Divorced	0	1	0		
Not reported	1	3	0		
Breast cancer diagnosis					
Left	13	4			
Right	4	2			
Bilateral	1	0			
Disease stage ^a					
I	1	0			
II	11	3			
III	6	3			
Chemotherapy regimen					
AC/paclitaxel	16	6			
EC/paclitaxel	1	0			
AC/docetaxel	1	0			
Adjuvant or neo-adjuvant					
Adjuvant	11	3			
Neo-adjuvant	7	3			
Type of surgery					
Breast-conserving surgery	9	3			
Mastectomy	9	3			
Axillary lymph node dissection					
+	16	5			
–	2	1			

^a Disease stage which was determined prior to chemotherapy

Discussion

This is an observational study including two time points [baseline (T1), 4 weeks after chemotherapy (T2)] in 18 breast cancer patients undergoing chemotherapy versus 20 healthy controls. Neuropsychological testing was performed at T1 and T2, and psychological functioning was

assessed with the HADS. We found baseline differences between patients and healthy controls in digit symbol and verbal fluency, in which patients had lower performance than healthy controls. No difference in change scores of neuropsychological tests was found between chemotherapy and control groups. However, only in the chemotherapy group, change in performance on logical memory and

Table 2 Neuropsychological and psychological characteristics of participants and non-participants, and interval between T1 and T2

	Chemotherapy group participants	Chemotherapy group non-participants	Control group	Chemo participants vs. control participants	
				<i>t</i> score	<i>p</i> value
Examined for cognitive assessment at	T1 and T2 (<i>n</i> = 18)	T1 only (<i>n</i> = 6)	T1 and T2 (<i>n</i> = 20)		
Estimated IQ (partial WAIS-R)					
Mean (SD)	113.4 (14.1)	111.2 (9.3)	116.5 (12.0)	0.722	0.475
Range	83–140	102–125	90–136		
Mood [mean (SD)]					
Depression at T1 (HADS-D score)	5.1 (2.9)	4.5 (2.1)	3.8 (2.6)	1.395	0.171
Anxiety at T1 (HADS-A score)	6.4 (2.3) ^a	7.0 (3.7)	4.1 (3.1)	2.451	0.019
Depression at T2 (HADS-D score)	4.2 (2.5) ^a		3.3 (1.9)	1.189	0.242
Anxiety at T2 (HADS-A score)	5.3 (2.5)		3.8 (2.8)	1.787	0.082
Interval between T1 and T2 (in days)					
Mean (SD)	263.6 (32.1)		275.3 (28.9)	−1.189	0.242
Range	217–345		224–356		
Median	257		269		

^a Missing data from one patient**Table 3** Comparison of change scores for each neuropsychological test between the chemotherapy (Chemo) and control (Healthy) groups

Domain	Test name	Chemo (<i>n</i> = 18)	Healthy (<i>n</i> = 20)	<i>t</i> score	<i>p</i> value	Effect size (<i>r</i>)
Verbal memory	Logical memory I	2.3 (3.4)	1.0 (4.0)	1.094	0.281	0.179
	Verbal paired associates I	0.9 (2.8)	1.5 (2.9)	−0.604	0.550	0.100
	Verbal memory (total)	5.4 (8.6)	3.4 (8.9)	0.733	0.468	0.121
Visual memory	Visual reproduction I	0.8 (3.7)	−0.3 (2.7)	1.084	0.286	0.178
Delayed recall	Logical memory II	2.5 (3.3)	1.7 (6.1)	0.539	0.594	0.089
	Verbal paired associates II	0.1 (1.3)	0.4 (1.1)	−0.758	0.454	0.125
	Visual reproduction II	0.6 (4.0)	−0.4 (4.8)	0.658	0.515	0.109
	Delayed recall (total)	3.2 (6.2)	2.0 (8.2)	0.512	0.612	0.085
Concentration	Digit span	−0.1 (2.1)	0.8 (2.3)	−1.200	0.238	0.196
Processing function	Digit symbol	0.4 (8.6)	3.4 (6.2)	−1.227	0.228	0.200
Visuospatial function	Block design	1.3 (4.5)	1.0 (4.4)	0.228	0.821	0.038
Executive function	Verbal fluency test-letter	−1.6 (6.3)	0.7 (6.3)	−1.084	0.286	0.178
	Verbal fluency test-category	0.9 (6.1)	−0.2 (6.5)	0.533	0.597	0.088
	KWCST-CA ^a	0.1 (1.8)	0.2 (2.1)	−0.127	0.900	0.022
	KWCST-PEN ^a	0.5 (2.8)	−0.7 (5.8)	0.802	0.428	0.135
	KWCST-PEM ^a	−0.1 (1.7)	−0.5 (4.7)	0.365	0.717	0.062

KWCST Keio version Wisconsin Card Sorting Test, CA categories achieved, PEN perseverative errors of the Nelson type, PEM perseverative errors of the Milner type

^a Chemotherapy group, *n* = 17

verbal fluency was negatively correlated with change in depression, and change in anxiety was negatively correlated with change in performance on verbal paired associates and digit symbol and positively correlated with change in performance on WCST. Moreover, a multiple regression analysis with group, change in anxiety, and change in depression as independent variables and change scores of individual neuropsychological tests as dependent

variables found only an association of change in anxiety with digit symbol performance. However, some significant results did not have sufficient observed power despite Cohen's recommended power of ≥ 0.8 [31].

No differences in change scores between T1 and T2 for each neuropsychological test between patients and healthy controls suggested that chemotherapy is not associated with cognitive side effects. On the other hand, regarding

Table 4 Comparison of change scores for HADS-D and HADS-A between the chemotherapy (Chemo) and control (Healthy) groups

	Chemo (<i>n</i> = 17) ^a	Healthy (<i>n</i> = 20)	<i>t</i> score	<i>p</i> value	Effect size (<i>r</i>)
HADS-D score	−1.1 (2.2)	−0.5 (1.8)	−0.866	0.392	0.145
HADS-A score	−1.1 (2.7)	−0.3 (2.5)	−0.891	0.379	0.149

HADS Hospital Anxiety and Depression Scale, *D* depression, *A* anxiety

^a Data are missing for one patient's HADS-A score at T1 and HADS-D score at T2

Table 5 Pearson's correlation coefficients for changes in depressed mood, anxiety, and cognitive function (T2–T1)

Domain	Test name	Change in HADS-D score (T2–T1)		Change in HADS-A score (T2–T1)	
		Chemo (<i>n</i> = 17 ^a)	Healthy (<i>n</i> = 20)	Chemo (<i>n</i> = 17 ^a)	Healthy (<i>n</i> = 20)
Change of each neuropsychological test score (T2–T1)	Verbal memory	Logical memory I	−0.766 ¹ , ***	−0.094	0.001
		Verbal paired associates I	−0.427	0.088	−0.263
		Verbal memory	−0.758 ² , ***	−0.055	−0.083
	Visual memory	Visual reproduction I	0.052	−0.164	−0.262
		Delayed recall	−0.498 ³ , *	−0.153	−0.073
		Logical memory II	0.033	0.151	−0.570 ⁴ , *
	Concentration	Verbal paired associates II	−0.248	−0.025	−0.081
		Visual reproduction II	−0.405	−0.089	−0.334
		Delayed recall	−0.198	−0.020	−0.314
	Processing function	Digit span	0.200	−0.380	−0.503 ⁵ , *
		Digit symbol	−0.199	−0.044	−0.293
		Block design	−0.168	−0.017	−0.276
	Visuospatial function	VFT-letter	−0.666 ⁶ , **	−0.129	−0.170
		VFT-category	−0.320	−0.043	0.516 ⁷ , *
		KWCST-CA ^b	0.137	−0.296	0.112
		KWCST-PEN ^b	0.176	−0.309	0.251
		KWCST-PEM ^b			−0.168

Chemo chemotherapy group, Healthy control group, T1 time 1 (prior to chemotherapy), T2 time 2 (1 month after chemotherapy), HADS Hospital Anxiety and Depression Scale, *D* depression, *A* anxiety, VFT verbal fluency test, KWCST Keio version Wisconsin Card Sorting Test, CA categories achieved, PEN perseverative errors of the Nelson type, PEM perseverative errors of the Milner type

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

^{1–7} Observed power values ($1 - \beta$) for significant results ($\alpha = 0.05$) were as follows: ¹0.97, ²0.97, ³0.56, ⁴0.71, ⁵0.57, ⁶0.87, ⁷0.56

^a Data are missing for one patient's HADS-A score at T1 and HADS-D score at T2

^b Missing data for one patient

baseline data of neuropsychological tests, mean raw scores at T1 of digit symbol and VFTs (letter and category) for the chemotherapy group were lower than the control group. Furthermore, HADS-A scores at T1 were significantly higher in the chemotherapy group than in the control group. These results might indicate that high anxiety experienced by patients at T1 makes their cognitive function during the first test appear lower, thereby making the difference between T1 and T2 appear larger. There is a possibility that chemotherapy actually induced cognitive change in patients in the present study, but apparent improvement in performance was observed between T1

and T2 because the difference appeared as large as that in healthy controls. Our data were not extensive enough to apply multivariate analysis, and further investigation will be necessary.

As for VFTs, previous studies reported that breast cancer patients performed significantly poorer compared to healthy controls on the verbal fluency test before initiating adjuvant chemotherapy [32], and significantly lower compared to normalized data at 1 month post-chemotherapy [33]. Our findings support the results of these studies. Verbal fluency has been commonly viewed as a component of executive function; however, the multidimensionality of

cognitive processes such as auditory attention and understanding, short-term memory, mental flexibility, speeded mental processing, and long-term vocabulary storage are involved [34]. Detailed examination and analysis of characteristics of word production and errors, such as perseveration, irrelevant associations, repetition of identical or very similar responses, and slowed performance, will provide more information about cognitive function in patients.

Vearncombe et al. [35] suggested that psychological factors might increase vulnerability of breast cancer patients to cognitive impairment after chemotherapy; however, many prospective studies have generally found no significant association between psychological distress and objective cognitive performance [14, 16, 27]. In fact, large, prospective, multivariate analysis studies of the general population reported that increased depressive symptoms were associated with poor cognitive functioning and cognitive decline in multiple domains [36, 37]. Furthermore, anxiety was also associated with poor cognitive functioning under the condition of adjustment for co-morbid depression [38]. A unique aspect of our analysis was to use change scores for HADS instead of concurrent HADS scores. This aspect could explain the significant results obtained in correlation analyses between cognitive function and psychological distress in the chemotherapy group. In the significant multiple regression model of this study, the standard partial regression coefficients of change in anxiety score were statistically significant, but those for the group were nearly significant ($p = 0.071$). However, these results were underpowered. Further investigation will be needed to accumulate knowledge about the relationship between chemotherapy-induced cognitive function and psychological distress.

The present study revealed that impaired CA (KWCST) score was related to improved HADS-A score. WCST is found to be a component of executive function; however, the multidimensionality of cognitive processes is involved. This result was contrary to our hypothesis and requires further investigation.

Many studies reported that women tend to express a subjective feeling of memory impairment and difficulty in concentration during and after chemotherapy [39, 40]; however, no relationship was found between neuropsychologically assessed cognitive impairment and self-reported cognitive difficulties in breast cancer survivors [2, 6, 15, 41, 42]. Other studies have suggested that there was a relationship between self-reported cognitive difficulties and psychological distress [18, 24, 42]. It is possible that perceptions of cognitive difficulties are correlated with psychological distress in cancer patients receiving chemotherapy. A recent prospective study found that the use of tamoxifen, but not aromatase inhibitor, had some effects on

cognitive function in breast cancer patients [43], whereas another study found no significant effects of tamoxifen or aromatase inhibitor [44]. The relationship between cognitive change and diagnosis/treatment of breast cancer is thus very complicated. Even if cognitive impairment were not detected by cognitive tests, perceived cognitive impairment could strongly affect perceived quality of life in patients.

When interpreting the findings of this study, several limitations must be considered. First, the number of participants in this study was small; therefore, neuropsychological test results and scores of the self-reported questionnaire may not fully represent the general population of women diagnosed/undiagnosed with breast cancer. As discussed above, our results were underpowered. Second, although the education level between patients and healthy controls was very different, there was no significant difference between the two groups in estimated IQ. In future studies, education level should be matched in the two groups. Third, attrition rate of participants, especially in the chemotherapy group, might have produced bias in the study results. Patients who dropped out might have experienced cognitive impairment due to chemotherapy, and if so, they may have refused to participate in the study as they felt a burden to complete the test set. Fourth, it is possible that the neuropsychological tests used in this study were not sensitive enough to detect subtle cognitive impairment. Furthermore, given the large number of comparisons performed, some of the significant associations could have arisen by chance. The issue of mass significance should be considered when interpreting results from this study. It may have been prudent to accept variables as important only if they had a p value less than 0.01.

It is challenging to clarify the influences of chemotherapy on neuropsychological function because therapeutic strategies for breast cancer and variables of each participant, including comparison group, are diverse and complex. According to qualitative research, the proportion of patients receiving chemotherapy who reported that they were experiencing problems with cognitive function after chemotherapy was far greater than that identified as having objective cognitive impairment [42]. In fact, according to narratives of survivors who received chemotherapy, some breast cancer survivors suffered from cognitive impairment and reported diminished quality of life and daily functioning [45–47]. It is important to be aware of the possible effects of cancer diagnosis itself on cancer therapies to support patients and improve their quality of life. Such information could help women understand their cognitive difficulties better, as they can feel and adjust to them more effectively. Studies of the effects of chemotherapy and other therapies on cognitive function and mood are important to understand the impact on quality of life in breast cancer survivors.

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